

Recap: Repetition Structures

while structure: loop while the condition is True.

Syntax: while condition:

```

[statement
 statement]

```

while block

True

False = out of the loop

Ex

```

n = 0
while n <= 5:
    print(n)
    n += 2
print("Done")

```

Short-hand
n = n + 2

Outside of the sum

Output: 0
2
4
Done

#1
n <= 5 → 0 <= 5
print(0)
n += 2 = 0 + 2 = 2

#2
n <= 5 → 2 <= 5
print(2)
n += 2 = 2 + 2 = 4

#3
n <= 5 → 4 <= 5
print(4)
n += 2 = 4 + 2 = 6

#4
n <= 5 → 6 <= 5 → False → Out!

Q19: Find the sum from user's inputs.
Stop when the user types -999999.

Ver 1. without break

```

num = int(input()) # num != 0
total = 0
while num != -999999:
    total += num
    num = int(input())
print(total)

```

Ver 2. with break

```

total = 0
while True:
    num = int(input())
    if num == -999999:
        break
    else:
        total += num
print(total)

```

idea is when num == -999999 break

only way to get out

Q20 Input Validation *

Ver 1. without break

```

hour = int(input())
while hour < 0:
    print("Invalid")
    hour = int(input())

```

loop while it's invalid

Ver 2. with break

```

while True:
    hour = int(input())
    if hour >= 0:
        break

```

if it's valid break

```

while True:
    print("Invalid")
    hour = int(input())
    rate = int(input())
    while rate < 0:
        print("Invalid")
        rate = int(input())
    salary = hour * rate
    print(salary)

```

```

if hour >= 0:
    break
print("Invalid")
while True:
    rate = int(input())
    if rate >= 0:
        break
    print("Invalid")
salary = hour * rate
print(salary)

```

For loop: for loop through the list of items

Syntax: for var in [item1, item2, item3]:

statement
statement

* the number of items = num loop

Ex for x in [10, 20, 30]:

```

    print("Hi")
    print(x**2)

```

Output: Hi }
100 }
Hi } 3 loops
400 }
Hi }
900 }

#1
x=10
print("Hi")
print(100)

#2
x=20
print("Hi")
print(400)

#3
x=30
print("Hi")
print(900)

Ex User wants to enter n values for you to do the calculation.

- User will first enter n.
- Then user will enter n values or n**k values

Ex BMI for n people. $BMI = w/h^{**2}$

n = int(input("Number of people"))

for i in range(0, n):

$n = \text{int}(\text{input}(\dots))$
 $[0, 1, \dots, n-1]$
 $= n \text{ items}$
 for i in $\text{range}(0, n, 1)$:
 $\text{weight} = \text{float}(\text{input}())$
 $\text{height} = \text{float}(\text{input}())$
 $\text{print}(f'\text{BMI \#}{i+1} = \{\text{weight}/\text{height}^2\})$

loop n times

range() function

↳ generate a sequence of numbers

1) $\text{range}(\text{start}, \text{end}, \text{step})$

ex: $\text{range}(0, 5, 1) \rightarrow [0, 1, 2, 3, 4]$

$\text{range}(7, 1, -3) \rightarrow [7, 4]$

2) $\text{range}(\text{start}, \text{end})$ # $\text{step} = 1$

ex: $\text{range}(0, 5) \rightarrow [0, 1, 2, 3, 4]$

3) $\text{range}(\text{end})$ # $\text{start} = 0, \text{step} = 1$

ex: $\text{range}(5) \rightarrow [0, 1, 2, 3, 4]$

Ex: $\text{total} = 0$
 for n in $\text{range}(4)$:
 $\text{total} += n$
 $\text{print}(\text{total}) \rightarrow [6]$

$[0, 1, 2, 3]$

Continue statement

↳ when you encounter continue, you goto the next loop

Output: Hi
 Hi
 Hi
 Hi

for x in $[10, 20, 30, 40]$
 $\text{print}('Hi')$
 continue
 $\text{print}(x)$

List Part 1

... collection of items

List Tutorial

List is a data type - a collection of items that we can access them by position (index)

Syntax: [item 1, item 2, item 3]
be any data type (int, float, string, ...)

Ex my_list = [1, 2, "A", "B", True]
print(my_list) → [1, 2, "A", "B", True]

Indexing: access an item in the list by referring to its index (position)

⊛ index starts at 0!! ⊛

my_list = [10, 20, 30, 40]
index 0 1 2 3

Syntax: my_list[index]

ex print(my_list[0]) → 10
a number

x = my_list[1] + my_list[2]
20 + 30 = 50
print(x) → 50

Negative Index: the last item is -1 and you count down from the back.

my_list = [10, 20, 30, 40]
-4 -3 -2 -1

print(my_list[-1]) → 40

print(f"{my_list[2] + my_list[3]}") → 50
30 + 20 = 50

Update list: update at a specific index.

Update list : update on a specific index.

Syntax: $\text{my_list}[\text{index}] = \text{value}$

EX $\text{list1} = [10, 20, 30, 40]$
 $\text{print}(\text{list1}) \rightarrow [10, 20, 30, 40]$
 $\text{print}(\text{list1}[0]) \rightarrow 10$
 $\text{list1}[0] = 5$
 $\text{print}(\text{list1}) \rightarrow [5, 20, 30, 40]$
 $\text{list1}[2] = \text{list1}[0] + \text{list1}[1] = 25$
 $\text{print}(\text{list1}) \rightarrow [5, 20, 25, 40]$

* Error: index out of range *

↳ you try to access index that not in the list
 $\text{print}(\text{list1}[20])$

try to access index 20 but not in the list $\rightarrow$ Error!

len() function

↳ $\text{len}(\text{list})$ tells you how many items in the list.

$\text{list2} = [10, 20, 30, 40]$
 $\text{print}(\text{len}(\text{list2})) \rightarrow 4$

$x = \text{len}([1, 2, 5, 9, 16]) = 5$
 $\text{print}(x) \rightarrow 5$

List Operators

. + : concatenate two lists together

$[1, 2] + [3, 4] \rightarrow [1, 2, 3, 4]$

$["A", 1] + ["B", 3] \rightarrow ["A", 1, "B", 3]$

$["A", 1] + ["B", 3] \rightarrow ["A", 1, "B", 3]$

* : Syntax: list * n
↳ repeat the list n times

$[1, 2] * 3 \rightarrow [1, 2, 1, 2, 1, 2]$
3 times

Loop through list

① numbers = [10, 20, 30, 40]
for num in numbers:
 print(num)

Output: 10
20
30
40

~~Ex~~ numbers = [10, 20, 30, 40]
for x in [0, 1]:
 print(numbers[x])

#1 x=0
print(numbers[0]) → 10
#2 x=1
print(numbers[1]) → 20

② loop through the list of indexes

numbers = [10, 20, 30, 40] → [0, 1, 2, 3]
nlen = len(numbers) = 4
indexes = range(nlen) → [0, 1, 2, 3]

range(4) → numbers indexes

for i in indexes:
 print(numbers[i])

Output 10 numbers[0]
20 numbers[1]
30 numbers[2]
40 numbers[3]

#1 i=0
print(numbers[0])
#2 i=1
...

numbers = [10, 20, 30, 40] range(4) → [0, 1, 2, 3]

for i in range(len(numbers)):
 print(numbers[i])

~~Ex~~ update the list

Ex Update the list

$[1, 2, 3, 4] \rightarrow [1, 4, 9, 16]$

nums = [1, 2, 3, 4]

Update syntax:
list[index] = new value

for i in range(len(nums)):

nums[i] = nums[i] ** 2

print(nums) $\rightarrow [1, 4, 9, 16]$

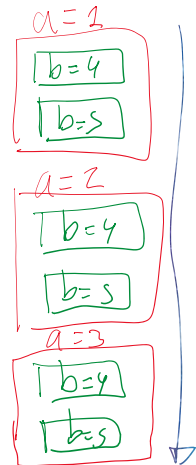
Nested Loops: loop can be inside of another loop

```
for a in [4, 2, 3]:  
    for b in [4, 5]:  
        print(a, b)
```

loop string

Output:

1	4
1	5
2	4
2	5
3	4
3	5



total=0

[0, 1, 2]

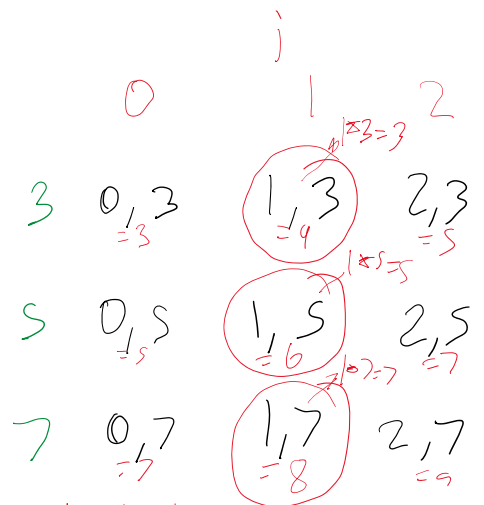
for i in range(0, 3):

for j in [3, 5, 7]:

if (i+j) % 2 == 0:

total += (i*j)

print(total) $\rightarrow 15$



total += 3 + 5 + 7 = 15